Cortical Specifi cation and Neuronal Migration 27 In general, these theories can be simplifi ed as mechanisms 1
operating at the level of the precursor cells within each neocortical area and to those infl uenced by migrating and differentiating neurons and to afferent cortical projections. For example, both of the expansion hypotheses rest on differential cell pro- duction, either in the radial domain (i.e., laminar differences) or the tangential domain (i.e., cortical column number). Both, therefore, suggest that isolated programs of neurogenesis along the protomap-specifi ed areas yield the result- ing pattern of cortical folding most appreciated after birth. Incidentally, while the axonal tension hypothesis primarily focuses on forces generated by dif- ferentiated neurons, it also invokes the area-specifi c projection and targeting patterns that must be conferred to the neocortical layers upon their generation from the underlying precursor cells. Thus, all theories for gyrifi cation require a signifi cant role for precursor cells in establishing the regions that will eventu- ally undergo convolution. Recent studies have provided a role for precursor gene expression and its consequences on neurogenesis as a predictor of gyrifi cation. In particular, a key study identifi ed the cell-adhesion molecules FLRT1 and FLRT3 in regulat- ing area-specifi c formation of gyri and sulci (Del Toro et al. 2017). These two genes are upregulated in the lissencephalic mouse neocortical wall and down- regulated in the gyrencephalic ferret and human neocortical wall in regions of incipient gyrus formation. This study shows that perturbations to lower the levels of FLRT1/3 expression lead to faster neuronal migration rates and to clusters of neurons expressing a similar level of these adhesion molecules. The increased radial and tangential pressure caused by these clustered neurons is postulated to lead to localized formation of a gyrus, even in the normally smooth mouse neocortex. Another gene recently identifi ed in the germinal zones that plays a role in neocortical folding is Trnp1, which encodes a nuclear protein potentially involved in chromatin state (Stahl et al. 2013). Knockdown of Trnp1 alters the pattern of cell division in the VZ, causing overproduc- tion of bIPCs and bRGCs. The hypothesis of this study is that increases in the numbers of resulting neurons, and their tangential spread afforded by the increased number of bRGC fi bers, result in gyrus formation in the overlying neocortical sheet. A study by de Juan Romero et al. (2015) offers perhaps the most convincing argument for a link between differential gene regulation in the germinal zones, expansion of bRGCs, and formation of overlying cortical convolutions. In this study, regions of the ferret neocortical wall, which later develop either a gyrus or a sulcus, were isolated at a stage in early develop- ment prior to the formation of these folds. Upon microarray profi ling, many hundreds of genes were differentially expressed between these two areas, and clear expression of these genes in the oSVZ in the future gyrus site was con- trasted to the lack of their expression in the oSVZ of the neighboring future sulcus site. In addition, a human RGC-specifi c gene, ARGAP11B, was shown to increase basal precursor proliferation and cortical folding when introduced into the developing mouse germinal zones (Florio et al. 2015). From “The Neocortex,” edited by W. Singer, T. J. Sejnowski and P. Rakic. Strüngmann Forum Reports, vol. 27, J. R. Lupp, series editor. Cambridge, MA: MIT Press. ISBN 978-0-262-04324-3